



M.SC. DATA ANALYTICS & TECHNOLOGIES

DAT7301- DATA ANALYSIS AND VISUALISATION

ASSESSMENT 2 - IMPLEMENTATION OF STATISTICAL ANALYSIS PLAN

Student Name: Victor Ene

Student Number: 2318217

Module Tutor: Dr Harinda Eugene

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Abstract

This report presents a detailed statistical analysis of UK seafood production and trade patterns from 2009 to 2020. Data were sourced from the Marine Management Organisation (MMO), HM Revenue and Customs (HMRC) and Centre for Environment, Fisheries and Aquaculture Science (CEFAS). The analysis included data cleaning, exploratory data analysis, correlation testing and time-series forecasting. Findings revealed a persistent seafood trade deficit and no strong correlation between domestic production and trade volumes. Forecasts using ARIMA and Holt's Linear ETS models suggest stable or modestly increasing production to 2025 with growing uncertainty. Limitations include the short timeframe, assumptions in imputations and models, and the exclusion of other trade factors. Insights from this analysis can support strategic decisions in seafood policy, supply chain management, and investment planning.

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1.0 Introduction

Data-driven analytical approaches and effective visual communication play an essential role in making strategic plans and efficiently running operations across different businesses and industries (Lucas and Smith, 2023). This report focuses on the UK seafood industry, examining the relationship between domestic production and international trade, species that most dominate the UK international trades and forecast seafood production. Understanding these dynamics is vital for policymakers, businesses, and stakeholders to inform food-security policy, quota negotiations and investments, optimise supply chains, forecast demand, and assess the impact of policies (Camm *et al.*, 2023).

Using R, this report details the implementation of a statistical analysis plan, covering data preparation, exploratory data analysis, advanced statistical techniques, visualisation and interpretation of findings and insights.

2.0 Data Collection and Preparation

The dataset for this analysis, covers the period from 2009 to 2020 and was obtained from multiple credible sources, including the Marine Management Organisation (MMO), HM Revenue and Customs (HMRC) and the Centre for Environment, Fisheries and Aquaculture Science (CEFAS) (Löfstedt *et al.*, 2025). The data preparation process involved several steps in R:

- i Loading Libraries: The necessary libraries, including tidyverse, forecast, ggplot2, dplyr, were loaded.
- ii Loading Data: The CSV file “UKSeafoodSupplyChain.csv” was read with read.csv, followed by an initial glimpse to check of data types and dimensions.
- iii Handling Missing Values: 47 records were missing production volumes. The data missing production volumes were imputed with the species-specific mean to ensure completeness for analysis (Field, 2018).
- iv Removing Unnecessary Columns: Columns such as “TemporalResolution”, “Flag”, “FlagDescription”, “DataSupplier”, “DataSet”, and “Units” were removed to focus on core analytical variables (calendar year, commodity, species, volume).
- v Unit Conversion: The Volume was converted from grams to tonnes to facilitate interpretation and comparison.
- vi Filtering Data: Commodity with records “Purchases” and “Consumption” were excluded as the study focuses on production, import and export only.
- vii Checking Duplicates: The dataset was checked for duplicate rows confirming no unintended repeated entries to ensure data integrity.

The resulting clean dataset, complete and free of extraneous columns formed the foundation for exploratory analysis, correlation testing and time-series forecasting.

	Commodity	Species	CalendarYear	Volume_tonne
1069	Imports	Prawn	2011	149000.00
1067	Imports	Prawn	2010	141000.00
1071	Imports	Prawn	2012	140000.00
1073	Imports	Prawn	2013	139000.00
1065	Imports	Prawn	2009	139000.00
1075	Imports	Prawn	2014	128000.00
1079	Imports	Prawn	2016	125000.00
1150	Exports	Salmon	2019	124000.00
1140	Exports	Salmon	2014	124000.00
1738	Production	Mackerel	2014	124000.00
1081	Imports	Prawn	2017	123000.00
1083	Imports	Prawn	2018	122000.00

Figure 1 Cleaned dataset sorted by descending volume

Source: Author.

3.0 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a method for examining datasets to summarise their key features, this is typically through visualisations and statistical techniques (Wickham, Cetinkaya-Rundel and Grolemund, 2023). An exploratory data analysis was performed to examine the data structure and uncover preliminary insights.

3.1 Summary Statistics

To gain initial insight into the structure and distribution of the cleaned UK seafood dataset (2009–2020) began with summary statistics, this step revealed that Production climbed from around 6200 tonnes in 2009 to a peak of 7550 tonnes in 2014, before easing back to about 6500 tonnes by 2020, while imports and exports displayed larger ranges.

CalendarYear <int>	Commodity <chr>	Mean_Volume <dbl>	Median_Volume <dbl>	SD_Volume <dbl>	Count <int>
2009	Exports	8746.462	1791.162	17888.87	54
2009	Imports	13674.183	2361.588	30429.40	56
2009	Production	6166.194	1086.576	14342.60	58
2010	Exports	9020.844	1751.865	19640.18	57
2010	Imports	12603.960	1691.649	29025.71	59
2010	Production	6407.978	1045.905	14356.74	58
2011	Exports	7488.423	1457.524	17125.53	58
2011	Imports	12917.065	1650.897	30303.63	59
2011	Production	6330.230	1179.361	15088.99	58
2012	Exports	7233.854	1250.700	17193.41	64

Figure 2 Summary Statistics

Source: Author.

Next, we plotted mean annual volume by commodity using ggplot2 for better visualisation (Wilke, 2019). Imports were at around 13700 tonnes in 2009, then fell steadily to approximately 11000 tonnes by 2015. A slight rebound to about 11500 tonnes occurred in 2016, after which imports stabilised near 11200 tonnes through to 2020. Exports showed greater volatility, rising from roughly 8700 tonnes in 2009 to 9000 tonnes in 2010, then declining to 7100 tonnes by 2013,

spiking again to about 8050 tonnes in 2014, and drifting downward to approximately 6700 tonnes by 2020.

3.2 Comparison of Imports and Exports

Total import and export volumes were aggregated by year and visualised using a grouped bar chart, allowing for a direct comparison of the UK seafood trade balance. Figure 4 presents annual totals of UK seafood imports and exports from 2009 to 2020. In every year imports exceeded exports, indicating a persistent net inflow of seafood into the UK. Imports peaked at around 800000 tonnes in 2012, while exports reached their high of approximately 520000 tonnes in 2010. The largest gap occurred in 2012 with 800000 tonnes imported versus 470000 tonnes exported, highlighting a trade deficit of roughly 330000 tonnes that year. This imbalance suggests that domestic production alone cannot meet home demand, emphasising UK reliance on international seafood supply and the importance of resilient trade networks (HRMC, 2025).

3.3 Dominant Import and Export Species

To identify the dominant species, total export and import volumes were summarised by species and commodity. The top 10 species for both exports and imports were identified and a horizontal bar plot in figure 5 further illustrated these dominant species, Bump charts were used to visualise the top 5 species in both imports and exports for each year, showing changes in dominance over the period.

Prawn dominates imports in the dataset, exceeding 1.6 million tonnes in total imports, highlighting the UK dependence on imported shellfish. Following prawn, tuna and cod with approximately 1.3 million tonnes and 1.2 million tonnes rank second and third, reflecting both consumer demand and domestic supply limitations. Notably, salmon approximately 0.9 million tonnes features

prominently in imports despite the UK robust farmed-salmon sector. Salmon leads UK seafood exports at 1.2 million tonnes, primarily supplying EU and North American markets. Mackerel approximately 0.9 million tonnes is second followed by Herring 0.5 million tonnes, reflecting its historic herring industry, which is now export-focused.

Figure 7 illustrates the cumulative domestic production volumes for the ten most-produced seafood species in the UK over 2009 to 2020. Mackerel and Salmon lead by a substantial margin, each exceeding 1000000 tonnes, Herring at roughly 680000 tonnes, ranks third. The next tier, Haddock, Crab, Cod and Nephrops represents a mix of demersal species and crustaceans, indicative of varied fishing pressures and market demand (Löfstedt *et al.*, 2025). Smaller but still significant volumes of Scallop, Monkfish and “Other marine fish” complete the top ten. This distribution highlights that UK production is dominated by a few high-yield species, with a long tail of lower-volume catches (CEFAS, 2015).

4.0 Application of Advanced Techniques

4.1 Correlation Between Seafood Production and Total Trade Volume

Correlation analysis was conducted to examine the relationship between the UK seafood production and its total trade volume (imports + exports). Pearson correlation coefficient was used to measure the linear relationship. A scatter plot of annual UK seafood production and total trade (imports + exports) for 2009–2020 showed the fitted line is essentially flat, Pearson's $r = -0.08$ ($p = 0.81$) indicating that production explains under 1% of the variation in trade and there is no statistically significant linear relationship. The wide confidence interval for the regression line emphasises the high uncertainty around the slight negative slope. Which indicates that annual fluctuations in UK seafood production and trade volumes appear to operate largely independently, suggesting that other factors (for example, market demand, pricing or regulation) drive trade more than domestic production. A correlation heatmap provided an overview of the correlations between production, imports, and exports.

4.2 Production vs Imports

The scatter of annual production against imports shows a very weak negative trend (Pearson's $r \approx -0.09$). The broad 95 % confidence band emphasises that any downward slope is indistinguishable from random variation. In practical terms, higher domestic landings have not noticeably reduced import volumes, suggesting that consumer demand for unproduced species (e.g. prawns, tuna) and other market factors govern import levels far more than local supply (MMO, 2022).

4.3 Production vs Exports

Plotting production against exports reveals a slight positive slope (Pearson's $r \approx +0.15$), yet the non-significant p-value indicate this association is too weak to be meaningful. Thus, increasing UK

landings has not reliably translated into greater exports. Export volumes appear driven primarily by external demand, quota arrangements and processing capacity rather than by production volume alone (MMO, 2022).

4.4 UK Seafood Production ARIMA Forecast

Time-series analysis was conducted using the Auto-Regressive Integrated Moving Average (ARIMA) modelling framework, a standard approach for forecasting data where historical patterns are expected to inform future values (Chatfield and Xing, 2019). The ARIMA model was automatically selected using the `auto.arima` function in R, ensuring optimal parameter choice. In the ARIMA forecast visualisation in figure 12, the blue markers represent the historical seafood production volumes from 2009 to 2020, while the red points correspond to ARIMA model forecasts from 2021 to 2025. The forecasted values remain constant at approximately 390000 tonnes annually. Unlike Holt's linear trend model, the ARIMA projection depicts a flat production path, indicating that historical volatility does not provide strong evidence of an upward or downward trend. This suggests the ARIMA model anticipates stability, but with considerable uncertainty as reflected in the 95 % confidence intervals (Hyndman and Athanasopoulos, 2021).

4.5 UK Seafood Production Holts Linear ETS Forecast

In Holts linear ETS forecast as well, the blue points mark the historical production from 2009 to 2020, while the red points show the five-year projections from 2021 to 2025. Holt's additive-trend model fits a gentle upward slope and production is forecasted to rise from around 385000 tonnes in 2021 to approximately 405000 tonnes by 2025, reflecting growing uncertainty the further we forecast (Box *et al.*, 2016).

5.0 Visualisation of insights

The analysis incorporated various visualisations created using the `ggplot2` package in R. These visualisations effectively communicate the key findings of the analysis, making it easier to understand the trends and relationships within the UK seafood industry (Hyman *et al.*, 2024).

5.1 Mean seafood volume by Year and Commodity

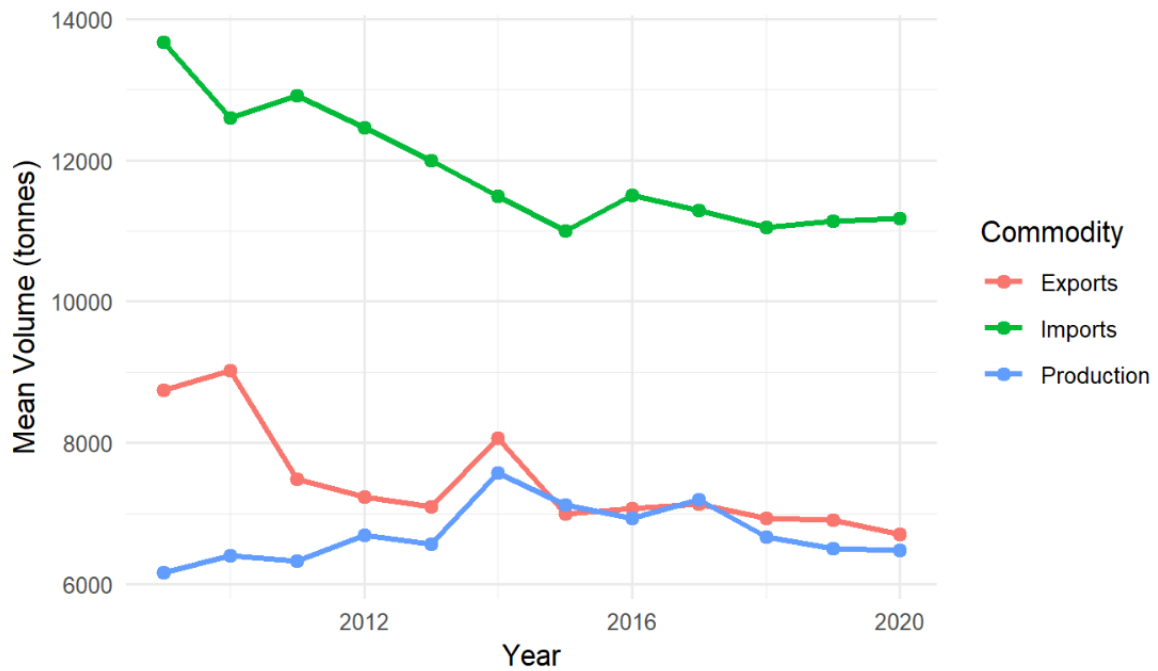


Figure 3 Mean seafood volume by Year and Commodity

Source: Author.

5.2 UK Seafood Import and Export

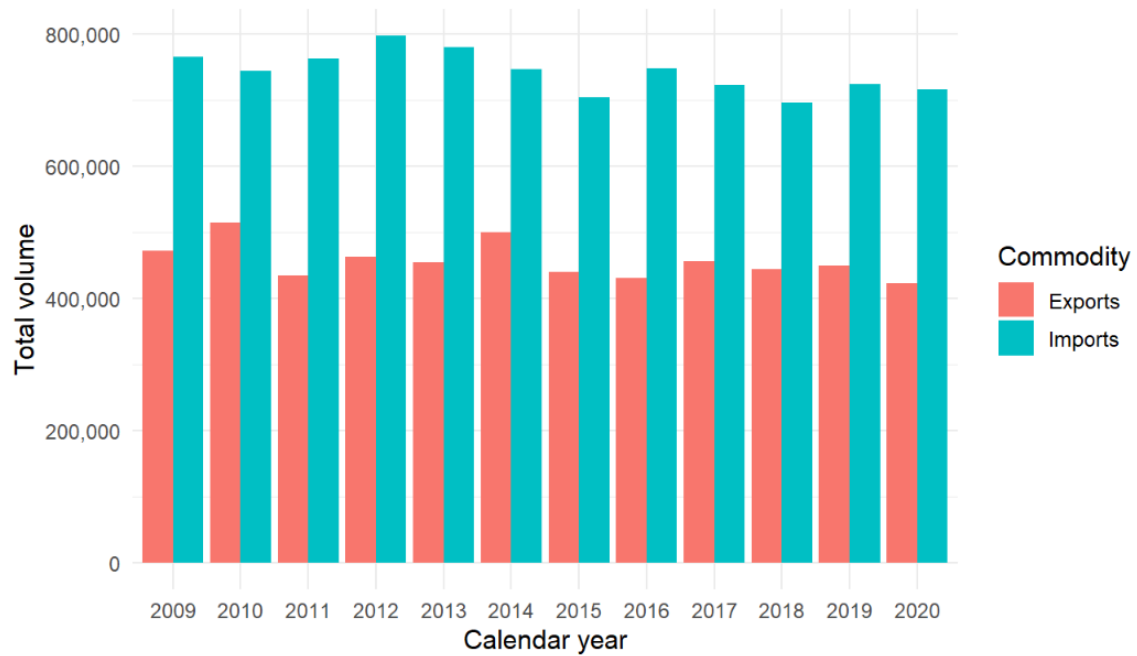


Figure 4 UK Seafood Import and Export

Source: Author.

5.3 Dominant seafood species

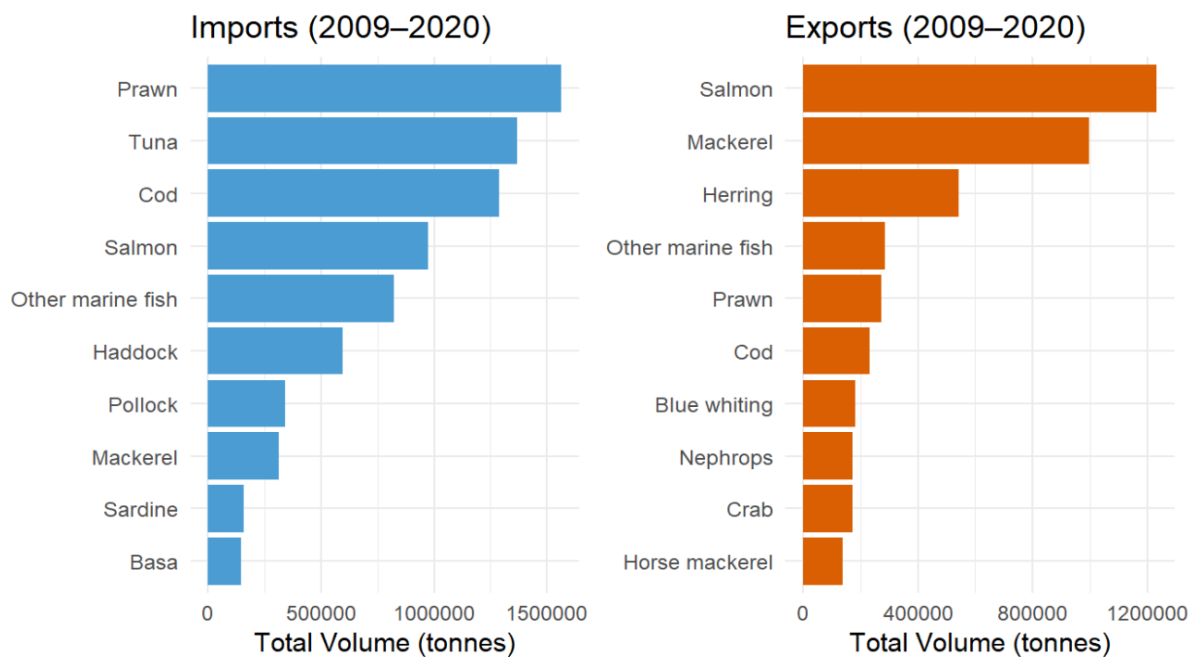


Figure 5 Dominant seafood species

Source: Author.

5.4 Top five Species by year

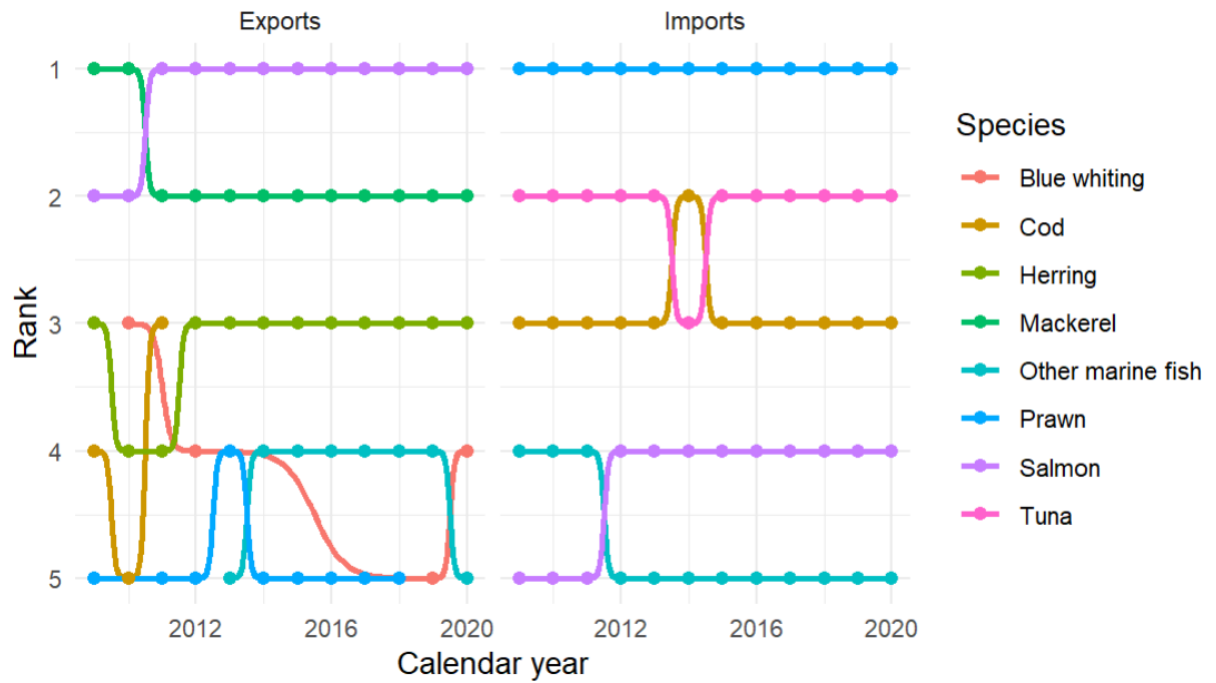


Figure 6 Top five Species by year

Source: Author.

5.5 Top 10 Species by production

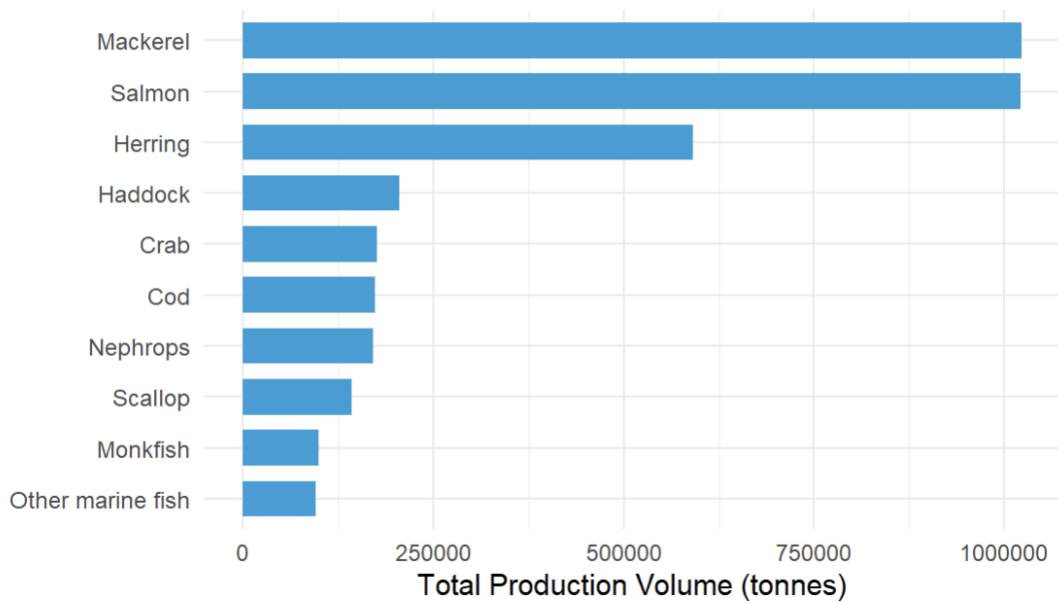


Figure 7 Top 10 Species by production

Source: Author.

5.6 Correlation between seafood production and trade

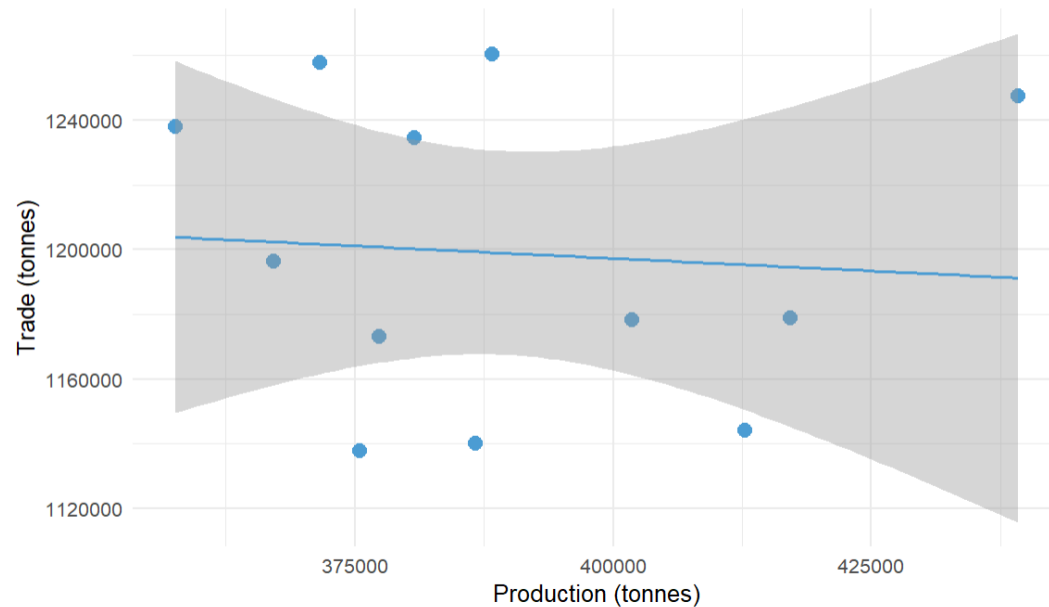


Figure 8 Correlation between seafood production and trade

Source: Author.

5.7 Correlation between seafood production and Imports

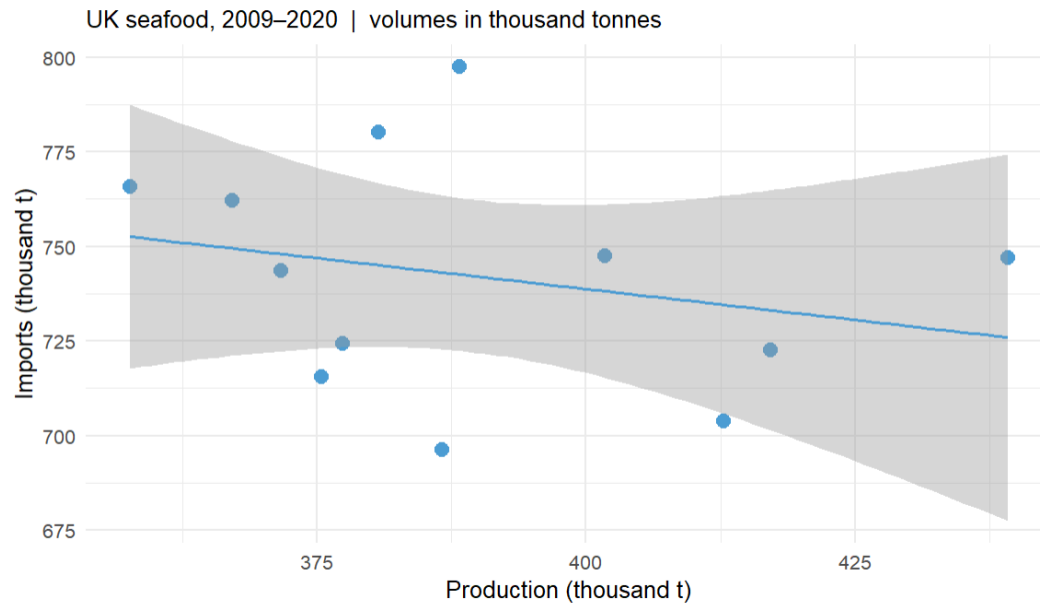


Figure 9 Correlation between seafood production and Imports

Source: Author.

5.8 Correlation between seafood production and Export



Figure 10 Correlation between seafood production and Exports

Source: Author.

5.9 Correlation Matrix: Production, Imports, Exports

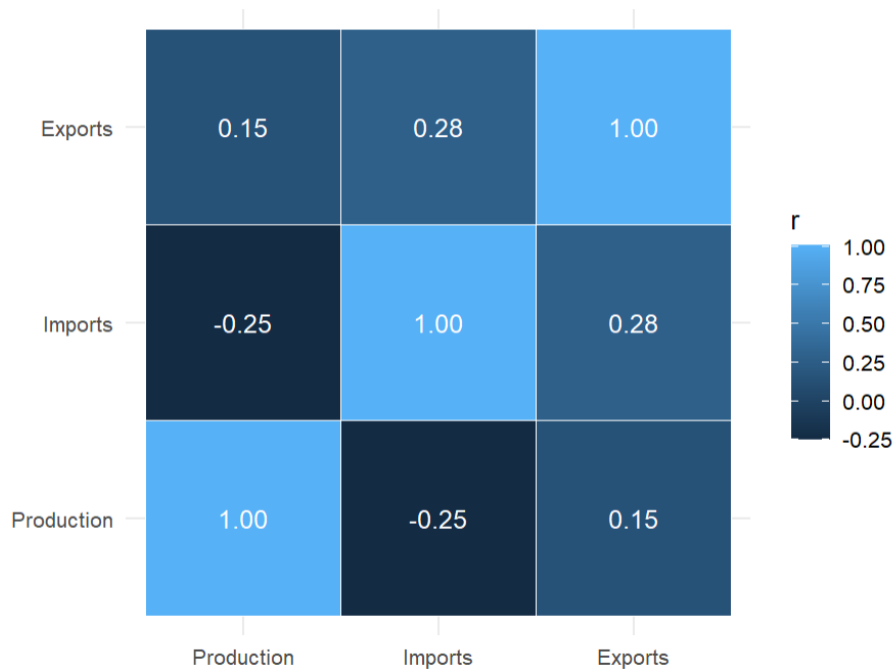


Figure 11 Correlation Matrix: Production, Imports, Exports

Source: Author.

5.10 ARIMA Forecast

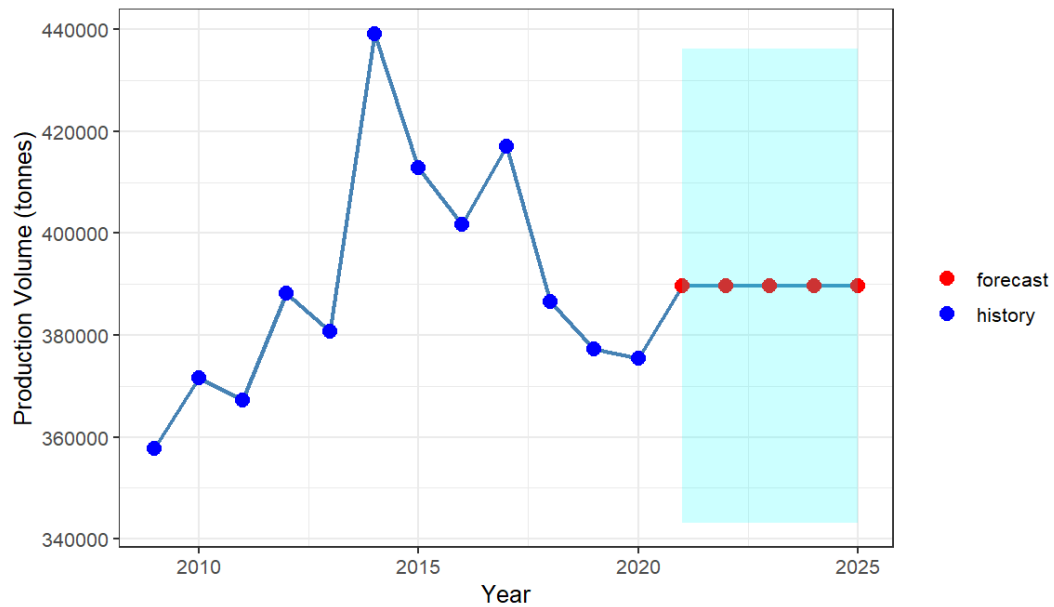


Figure 12 UK Seafood Production ARIMA Forecast

Source: Author.

5.11 ETS Forecast

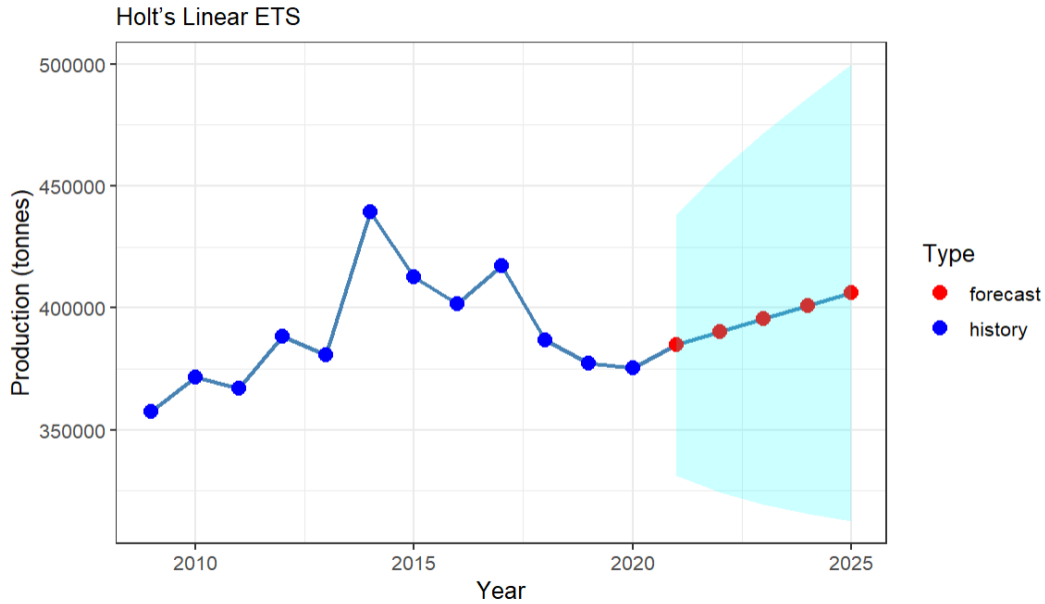


Figure 13 UK Seafood Production ETS Forecast

Source: Author.

6.0 Critical Interpretation

The exploratory data analysis (EDA) offered detailed insights into the structure and trends of UK seafood production and trade between 2009 and 2020. The identification of dominant species in both imports and exports, such as prawns, tuna, cod, and salmon, provides valuable information for shaping trade policies and targeting market strategies (Löfstedt et al., 2025).

The correlation analysis, revealed only weak and statistically insignificant relationships between domestic production and trade volumes. Production and imports demonstrated a very slight negative correlation, while production and exports showed a minor positive correlation, implying that other market factors such as international demand, trade agreements and environmental regulations likely have a stronger influence on trade volumes than domestic production levels alone (MMO, 2022; FAO, 2022).

Time-series forecasting using ARIMA and Holt's Linear ETS models provided valuable foresight into future seafood production trends. The ARIMA model projected a stable production without growth, suggesting that past volatility offers limited predictive power. While, Holt Linear ETS model forecasted a gentle upward trend in production, with increasing uncertainty over time, the accuracy of the time-series forecasts depends on the assumptions of the models and the stability of the historical trends (Chatfield and Xing, 2019).

Despite these valuable insights, the analysis faces several limitations, the dataset covers a relatively short period of 12 years and does not include post 2020 developments such as the effects of Brexit, COVID-19, or recent policy shifts. Secondly, while missing values were addressed using species specific mean imputation, this approach assumes that the missing data are random, which may not always be true (Zuur, Leno and Smith, 2007).

7.0 Conclusion

This study successfully implemented a statistical analysis plan to explore UK seafood production and trade between 2009 and 2020, using techniques including correlation analysis, time-series forecasting and comprehensive data visualisation. Key findings highlighted a persistent seafood trade deficit, dominated by large scale prawn and tuna imports, and significant exports of salmon and mackerel.

Forecasting with ARIMA and ETS models indicated stable or slightly increasing production trends over the next five years, but with considerable uncertainty. These forecasts, though insightful, must be interpreted cautiously given the limitations of historical data and the unpredictable nature of global seafood markets.

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Appendices

```
---
title: "Data Visualisation"
author: "Victor Ene"
date: "2025-04-20"
output: html_document
---

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

```{r}
install.packages("ggbump")
install.packages("patchwork")
install.packages("ggpmisc")
Load libraries
library(tidyverse)
library(forecast)
library(ggplot2)
library(dplyr)
library(ggbump)
library(broom)
library(patchwork) # to combine plots
library(tidyverse)
library(ggpmisc)
```

# Getting and understanding the Data

```{r}
getwd()

Load dataset
seafood_data <- read.csv("UKSeafoodSupplyChain.csv")

Summarise the dataset
summary(seafood_data)

sum(is.na(seafood_data))

Count NA's in each column
na_per_col <- colSums(is.na(seafood_data))
print(na_per_col)
```

# Data Preparation and Cleaning
```

```

```{r}
Identify rows with missing values in any column
missing_rows <- seafood_data[rowSums(is.na(seafood_data)) > 0,]

View the rows with missing values
print(missing_rows)
```

```{r}
Impute missing production volumes
seafood_filled <- seafood_data |>
 group_by(Species, Commodity) |>
 mutate(
 Volume = ifelse(
 is.na(Volume) & Commodity == "Production",
 mean(Volume, na.rm = TRUE),
 Volume
)
) |>
 ungroup() #Drops grouping so later operations would not be unintentionally
grouped.

sum(is.na(seafood_filled))
```

```{r}
#Remove unnecessary columns, convert to tonnes and drop Purchases and
Consumption
cleanseafood_data <- seafood_filled |> select(-TemporalResolution, -Flag, -
FlagDescription, -DataSupplier, -DataSet, -Units, -SpeciesType, -SACN)

cleanseafood_data <- cleanseafood_data |>
 mutate(Volume_tonne = Volume / 1e6) |> #grams to tonnes
 select(-Volume) |>
 filter(!Commodity %in% c("Purchases", "Consumption"))

head(cleanseafood_data)
```

```{r}
Check for duplicate rows
duplicates <- seafood_filled[duplicated(seafood_filled),]

print(duplicates)
```

# Exploratory Data Analysis (EDA)

```

```

```{r}
summary_stats <- cleanseafood_data |>
 group_by(CalendarYear, Commodity) |>
 summarise(Mean_Volume = mean(Volume_tonne),
 Median_Volume = median(Volume_tonne),
 SD_Volume = sd(Volume_tonne),
 Count = n(),
 .groups = "drop")

print(summary_stats)

ggplot(summary_stats, aes(x = CalendarYear, y = Mean_Volume, color =
Commodity)) +
 geom_line(linewidth = 1) +
 geom_point(size = 2) +
 labs(title = "Mean Seafood Volume by Year and Commodity",
 x = "Year", y = "Mean Volume (tonnes)",
 color = "Commodity") +
 theme_minimal() +
 theme(legend.position = "right")
```

```{r}
Summarise all trade volumes by Commodity and Year
all_trade <- cleanseafood_data |>
 group_by(Commodity, CalendarYear) |> # Grouping by Commodity and Calendar
Year
 summarise(Total_Volume = sum(Volume_tonne, na.rm = TRUE), .groups = "drop")

View the summarized data
print(all_trade)

Keep only Imports & Exports, then aggregate by year
trade_IE <- cleanseafood_data %>%
 filter(Commodity %in% c("Imports", "Exports")) %>%
 group_by(CalendarYear, Commodity) %>%
 summarise(Total_Volume = sum(Volume_tonne, na.rm = TRUE), .groups = "drop")

print(trade_IE)

ggplot(trade_IE,
 aes(x = factor(CalendarYear),
 y = Total_Volume,
 fill = Commodity)) +
 geom_col(position = "dodge") +
 scale_y_continuous(labels = comma) +
 labs(title = "UK Seafood Imports vs Exports by Year",
 x = "Calendar year",
 y = "Total volume",

```

```

 fill = "Commodity") +
 theme_minimal(base_size = 12) +
 theme(legend.position = "right")
...

Which species dominate the UK seafood exports and imports
```{r}
# Summarise total export and import volumes by species
species_trade <- cleanseafood_data |>
  group_by(Species, Commodity) |>
  summarise(Total_Volume = sum(Volume_tonne, na.rm = TRUE), .groups = "drop")

print(species_trade)

# Identify dominant species for imports and exports
dominant_exports <- species_trade |>
  filter(Commodity == "Exports") |>
  arrange(desc(Total_Volume)) |>
  head(10)

print(dominant_exports)

dominant_imports <- species_trade |>
  filter(Commodity == "Imports") |>
  arrange(desc(Total_Volume)) |>
  head(10)

print(dominant_imports)

# Top 10 Imports bar-plot
p_imports <- ggplot(dominant_imports, aes(x = reorder(Species, Total_Volume),
y = Total_Volume)) +
  geom_col(fill = "#4b9cd3") +
  coord_flip() +
  labs(title = "Imports (2009-2020)",
    x = NULL,
    y = "Total Volume (tonnes)") +
  theme_minimal(base_size = 12)

# Top 10 Exports bar-plot
p_exports <- ggplot(dominant_exports, aes(x = reorder(Species, Total_Volume),
y = Total_Volume)) +
  geom_col(fill = "#d95f02") +
  coord_flip() +
  labs(title = "Exports (2009-2020)",
    x = NULL,
    y = "Total Volume (tonnes)") +

```

```

    theme_minimal(base_size = 12)

# Display side by side
p_imports + p_exports +
  plot_annotation(
    title = "Dominant Seafood Species",
    theme = theme(plot.title = element_text(size = 14, face = "bold"))
  )
...

```{r}
#top 10 species by Production
dominant_production <- species_trade |>
 filter(Commodity == "Production") |>
 arrange(desc(Total_Volume)) |>
 slice_head(n = 10)

ggplot(dominant_production, aes(
 x = reorder(Species, Total_Volume),
 y = Total_Volume
)) +
 geom_col(fill = "#4b9cd3", width = 0.7) +
 coord_flip() +
 labs(
 title = "Top 10 Seafood Species by Production (2009-2020)",
 x = NULL,
 y = "Total Production Volume (tonnes)"
) +
 theme_minimal(base_size = 12)
...

Yearly top 5 species
```{r}
species_year <- cleansefood_data |>
  filter(Commodity %in% c("Imports", "Exports")) |>
  group_by(CalendarYear, Commodity, Species) |>
  summarise(Total_tonnes = sum(Volume_tonne, na.rm = TRUE),
    .groups = "drop") |>
  group_by(CalendarYear, Commodity) |>
  mutate(Rank = rank(-Total_tonnes, ties.method = "min")) |>
  ungroup()

#top five in every year
species_year_top <- species_year |>
  filter(Rank <= 5)

ggplot(species_year_top,
  aes(x = CalendarYear,

```

```

        y = Rank,
        colour = Species)) +
geom_bump(size = 1) +
geom_point(size = 2) +
scale_y_reverse(breaks = 1:5) + # 1 = top
facet_wrap(~ Commodity) +
labs(title = "Rank of Top-5 Seafood Species",
      x = "Calendar year",
      y = "Rank",
      colour = "Species") +
theme_minimal(base_size = 11) +
theme(legend.position = "right")

...

# Further analysis
##What is the correlation between seafood production and total trade volume

```{r}
aggregate to annual totals
annual_totals <- cleanseafood_data |>
 filter(Commodity %in% c("Production", "Imports", "Exports")) |>
 mutate(TradeFlag = if_else(Commodity == "Production", "Production",
"Trade")) |>
 group_by(CalendarYear, TradeFlag) |>
 summarise(Total_tonne = sum(Volume_tonne, na.rm = TRUE), .groups = "drop")
|>
 pivot_wider(names_from = TradeFlag, values_from = Total_tonne)

print(annual_totals)

correlation tests
pearson_res <- cor.test(annual_totals$Production,
 annual_totals$Trade,
 method = "pearson")

print(pearson_res)

spearman_res <- cor.test(annual_totals$Production,
 annual_totals$Trade,
 method = "spearman")

print(spearman_res)

corr_tbl <- bind_rows(
 tidy(pearson_res) |> mutate(Method = "Pearson"),
 tidy(spearman_res) |> mutate(Method = "Spearman")
) |>
 select(Method, estimate, p.value, conf.low, conf.high)

```

```

print(corr_tbl)
```

```{r}

#Pearson r for display
r_val <- cor(annual_totals$Production, annual_totals$Trade, method =
"pearson")

2. Scatter + regression + annotation
ggplot(annual_totals, aes(x = Production, y = Trade)) +
 geom_point(size = 3, colour = "#4b9cd3") +
 geom_smooth(method = "lm", se = TRUE, colour = "#4b9cd3", linewidth = 0.8)
+
 labs(
 title = "Correlation between UK Seafood Production and Trade (2009-
2020)",
 x = "Production (tonnes)",
 y = "Trade (tonnes)"
) +
 theme_minimal(base_size = 12)
```

```{r}

annual totals
annual_wide <- cleanseafood_data %>%
 filter(Commodity %in% c("Production", "Imports", "Exports")) %>%
 group_by(CalendarYear, Commodity) %>%
 summarise(Total_tonne = sum(Volume_tonne),
 .groups = "drop") %>%
 pivot_wider(names_from = Commodity,
 values_from = Total_tonne) %>%
 arrange(CalendarYear)

print(annual_wide)

Production vs Imports
r_imp <- cor(annual_wide$Production, annual_wide$Imports)

ggplot(annual_wide,
 aes(x = Production / 1e3, y = Imports / 1e3)) +

```

```

 geom_point(size = 3, colour = "#4b9cd3") +
 geom_smooth(method = "lm", se = TRUE, linewidth = 0.7, colour = "#4b9cd3")
+
 labs(title = "Production vs Imports",
 subtitle = "UK seafood, 2009-2020 | volumes in thousand tonnes",
 x = "Production (thousand t)",
 y = "Imports (thousand t)") +
 theme_minimal(base_size = 12)
...

```{r}
# Production vs Exports
r_exp <- cor(annual_wide$Production, annual_wide$Exports)

ggplot(annual_wide,
       aes(x = Production / 1e3, y = Exports / 1e3)) +
  geom_point(size = 3, colour = "#d95f02") +
  geom_smooth(method = "lm", se = TRUE, linewidth = 0.7, colour = "#d95f02")
+
  labs(title = "Production vs Exports",
        subtitle = "UK seafood, 2009-2020 | volumes in thousand tonnes",
        x = "Production (thousand t)",
        y = "Exports (thousand t)") +
  theme_minimal(base_size = 12)
...

```{r}
#Correlation heat-map for the three flows
corr_mat <- annual_wide %>%
 select(Production, Imports, Exports) %>%
 cor(use = "pairwise.complete.obs")

corr_long <- corr_mat %>%
 as_tibble(rownames = "Var1") %>%
 pivot_longer(-Var1, names_to = "Var2", values_to = "r")

ensure the axes keep the original order
corr_long <- corr_long %>%
 mutate(Var1 = factor(Var1, levels = colnames(corr_mat)),
 Var2 = factor(Var2, levels = colnames(corr_mat)))

ggplot(corr_long, aes(x = Var1, y = Var2, fill = r)) +
 geom_tile(colour = "white", linewidth = 0.4) +
 geom_text(aes(label = sprintf("%.2f", r)), colour = "white", size = 4) +
 labs(title = "Correlation Matrix: Production, Imports, Exports",
 x = NULL, y = NULL) +
 coord_fixed() +
 theme_minimal(base_size = 12)

```



```

...

#forecasting

```{r}

# annual production ts
production_data <- cleanseafood_data %>%
  filter(Commodity == "Production") %>%
  group_by(CalendarYear) %>%
  summarise(Total = sum(Volume_tonne, na.rm=TRUE), .groups="drop")

prod_ts <- ts(production_data$Total, start =
min(production_data$CalendarYear), frequency = 1)

# Fit ARIMA and forecast 5 years
arima_model <- auto.arima(prod_ts, stepwise=FALSE, approximation=FALSE)
arima_fc <- forecast(arima_model, h=5)

# combined history + forecast dataframe
forecast_df <- bind_rows(
  production_data %>%
    rename(year = CalendarYear) %>%
    mutate(type = "history"),

  tibble(
    year = (max(production_data$CalendarYear) +
1):(max(production_data$CalendarYear) + 5),
    Total = as.numeric(arima_fc$mean),
    lower = as.numeric(arima_fc$lower[,2]),
    upper = as.numeric(arima_fc$upper[,2]),
    type = "forecast"
  )
) %>%
  arrange(year)

#continuous line connecting history + forecast
ggplot(forecast_df, aes(x = year, y = Total, group = 1)) +
  geom_line(size = 1, colour = "steelblue") +
  geom_point(aes(colour = type), size = 3) +
  geom_ribbon(
    data = filter(forecast_df, type == "forecast"),
    aes(ymin = lower, ymax = upper),
    fill = "cyan", alpha = 0.2
  ) +
  scale_colour_manual(values = c(history = "blue", forecast = "red")) +

```

```

labs(
  title   = "UK Seafood Production Forecast (2021-2025) - ARIMA",
  x       = "Year",
  y       = "Production Volume (tonnes)",
  colour  = ""
) +
theme_bw(base_size = 12)
...

```{r}

annual production series
production_data <- cleanseafood_data %>%
 filter(Commodity == "Production") %>%
 group_by(CalendarYear) %>%
 summarise(Total = sum(Volume_tonne, na.rm = TRUE), .groups="drop")

prod_ts <- ts(production_data$Total,
 start = min(production_data$CalendarYear),
 frequency = 1)

#Holt's linear ETS (additive trend)
ets_model <- ets(prod_ts, model = "AAN")
ets_fc <- forecast(ets_model, h = 5)

history_prod_df <- tibble(
 Year = time(prod_ts),
 Value = as.numeric(prod_ts),
 Type = "history"
)

forecast_prod_df <- tibble(
 Year = as.integer(time(ets_fc$mean)),
 Value = as.numeric(ets_fc$mean),
 Lower = as.numeric(ets_fc$lower[,2]),
 Upper = as.numeric(ets_fc$upper[,2]),
 Type = "forecast"
)

combined_prod <- bind_rows(history_prod_df, forecast_prod_df)

ggplot(combined_prod, aes(x = Year, y = Value)) +
 geom_line(colour = "steelblue", size = 1) +
 geom_point(aes(colour = Type), size = 3) +
 geom_ribbon(data = filter(combined_prod, Type == "forecast"),
 aes(ymin = Lower, ymax = Upper),
 fill = "cyan", alpha = 0.2) +
 scale_colour_manual(values = c(history = "blue", forecast = "red")) +

```

```

labs(
 title = "UK Seafood Production Forecast (2021-2025)",
 subtitle = "Holt's Linear ETS",
 x = "Year",
 y = "Production (tonnes)"
) +
theme_bw(base_size = 12)

...

```{r}

#Compare AIC
aic_vals <- data.frame(
  Model = c("ARIMA", "ETS"),
  AIC    = c(AIC(arima_model), AIC(ets_model))
)
print(aic_vals)

#Compare accuracy (RMSE, MAE)
acc_arima <- accuracy(arima_model)
acc_ets    <- accuracy(ets_model)
acc_df     <- data.frame(
  Model = rep(c("ARIMA","ETS"), each=ncol(acc_arima)),
  Metric= colnames(acc_arima),
  ARIMA  = as.numeric(acc_arima[1,]),
  ETS    = as.numeric(acc_ets[1,])
)
print(acc_df)

fc_combined <- ts.union(
  ARIMA = arima_fc$mean,
  ETS    = ets_fc$mean
)

autoplot(prod_ts, series="Actual") +
  autolayer(fc_combined, series=c("ARIMA","ETS")) +
  labs(title="Production: Actual vs ARIMA & ETS Forecasts",
       x="Year", y="Production (tonnes)") +

scale_colour_manual(values=c("Actual"="black","ARIMA"="grey50","ETS"="blue"))
+
  theme_minimal()

```